

Quiz 9

I am very bad at darts. When I throw a dart and hit the dartboard, the locations where I hit are uniformly distributed on the board. In other words, given that I hit the board, the probability that I hit in a region of the board is equal to the fraction of the board that it covers. For example, the probability that I hit the left half of the board is 0.5.

1. Suppose my dart board is a disc with radius 1 foot. What is the probability that my dart hits within 0.5 feet of the center, given that I hit the board? To calculate this probability, sketch the board and shade the part of the board that is within 0.5 feet of the center. The probability is the fraction of the board's total area that you have shaded.

The region within 0.5 feet of the center is a disc of radius 0.5. The probability of my hitting within this disc is $(\text{area of disc of radius } 0.5) / (\text{area of the dartboard}) = \pi (0.5)^2 / \pi (1)^2 = 0.25$.

2. We write R for a random variable representing the distance from the center of the board where my dart hits (given that I hit the board). Calculate the distribution function F for R . Your formula for $F(x)$ must include cases for all values of x and not just for x between 0 and 1.

The region of the board where the distance to the center is at most x is a disc of radius x . The probability of my hitting within that disc is $(\text{area of disc of radius } x) / (\text{area of the whole dartboard}) = \frac{\pi x^2}{\pi (1)^2} = x^2$.

We thus have the following formula for F :

If $x < 0$, $F(x) = 0$ (since the value of R is at least 0).

If $0 \leq x \leq 1$, $F(x) = x^2$ (by the area calculation above).

If $x > 1$, $F(x) = 1$ (since you are given that I hit the dartboard).

3. In the previous part, you calculated the distribution function for R , which turns out to be a continuous random variable. Calculate the density function f for R . Your formula for $f(x)$ must include cases for all values of x and not just for x between 0 and 1.

The density function f is F' . We have the following formula:

If $x < 0$, $f(x) = 0$.

If $0 \leq x \leq 1$, $f(x) = 2x$.

If $x > 1$, $f(x) = 0$.

4. Calculate the median of R , $E[R]$, and $\text{Var}(R)$.

Median of R :

To find the median of any random variable, we solve “distribution function = 1/2.” In this case, a median must be between 0 and 1, and so we have

$$\begin{aligned} F(m) &= \frac{1}{2} \\ m^2 &= \frac{1}{2} \\ m &= \sqrt{\frac{1}{2}} \approx 0.707 \end{aligned}$$

$E[R]$:

To calculate the expected value of a random variable with density function f , we integrate $xf(x)dx$ over the range of values for the random variable. In this case, since f is 0 outside of $[0, 1]$, we have

$$\begin{aligned} E[R] &= \int_0^1 xf(x)dx \\ &= \int_0^1 x(2x)dx \\ &= \int_0^1 2x^2dx \\ &= \left[\frac{2}{3}x^3 \right]_{x=0}^{x=1} \\ &= \frac{2}{3} \end{aligned}$$

$\text{Var}(R)$:

As for any random variable, $\text{Var}(R) = E[R^2] - E[R]^2$. We calculate $E[R^2]$ using the law of the unconscious statistician:

$$\begin{aligned}
 E[R^2] &= \int_0^1 x^2 f(x) dx \\
 &= \int_0^1 2x^3 dx \\
 &= \left[\frac{2}{4} x^4 \right]_{x=0}^{x=1} \\
 &= \frac{1}{2}
 \end{aligned}$$

We thus have $\text{Var}(R) = E[R^2] - E[R]^2 = \frac{2}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{9}{18} - \frac{8}{18} = \frac{1}{18}$.